RAQIB · AI INSPECTION ROBOT

Build the robot

What to buy, every part pictured and explained, how to wire each sensor, and how to put the robot together, step by step.



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Final-year IoT project · UNISIRAJ

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Matches the Raqib software 2.3

Inside this guide

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The golden rule: wire one part, test it, then add the next. When something does not work, you know it is the part you just added. Every step ends with a check box list and a test command.

Wire colours used in every diagram

5 V · 3.3 V · GND · SDA · SCL · battery + · motor · PWM · signal

Use the same colours for your real jumper wires where you can: it makes mistakes easy to spot.

Practise first, for free: the Wokwi picture guide lets you wire most of these sensors in a browser simulator, and the phone app's *Try the demo robot* button shows what the finished robot does.

1 How the robot works

The Raspberry Pi 4 is the brain. It runs the robot program and also its own small server, so the phone app talks straight to the robot over Wi-Fi: no laptop. Two batteries keep the electronics and the motors apart, a hand-made **hub board** shares power and the I²C bus with the sensors, and each sensor group has one job.

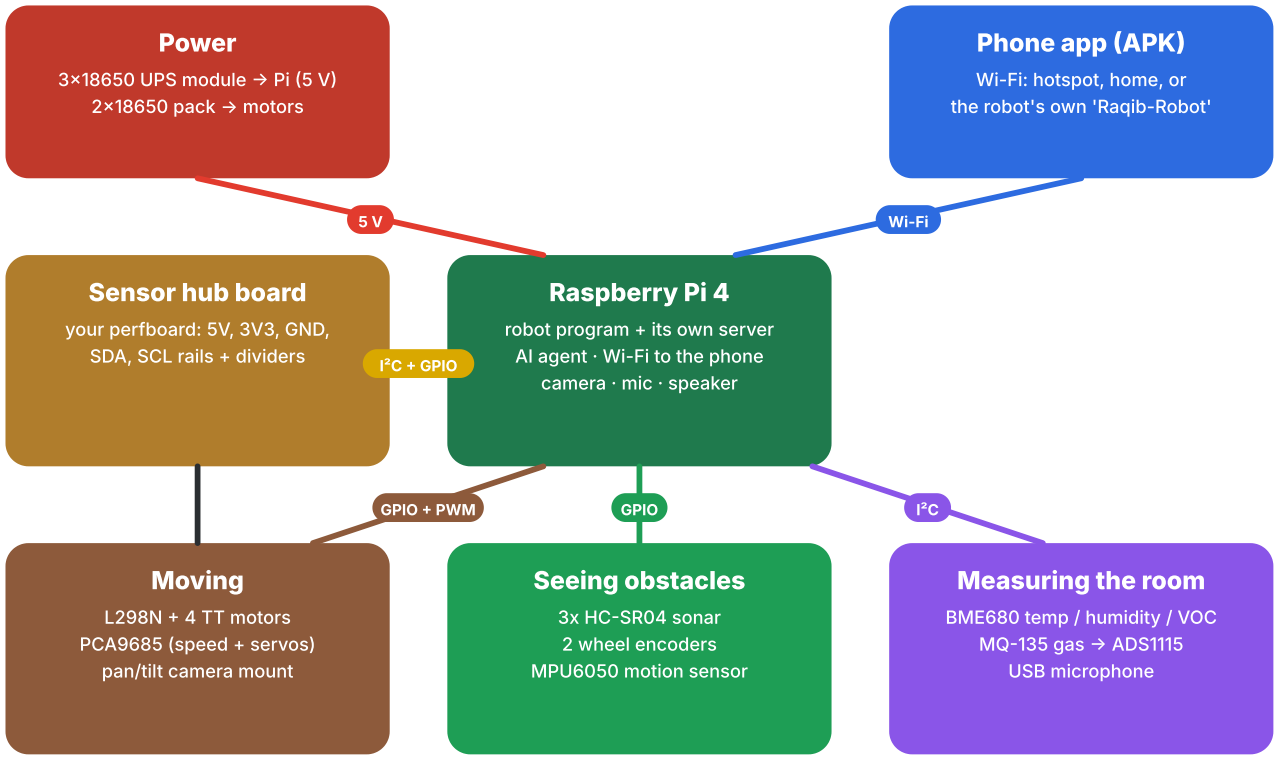


Figure 1 - the five groups of parts around the Raspberry Pi.

Two kinds of connection

I²C bus (2 wires, SDA + SCL): four boards share it, each with its own address - PCA9685 0x40, ADS1115 0x48, MPU6050 0x68, BME680 0x76, plus the UPS battery gauge.

GPIO pins: one wire per signal - motor direction, sonar trigger/echo and wheel encoders.

Two voltages

5 V for the sonars, gas heater, servos and amplifier.

3.3 V for everything that talks to the Pi. A Pi pin must **never** see 5 V: the voltage dividers on the hub board take care of the two 5 V outputs (sonar Echo and MQ-135).

2 Shopping list

Prices are **estimates in Malaysian ringgit** from typical online electronics shops (for example Cytron, or Shopee / Lazada electronics sellers) in 2026. Check current prices and stock before you buy.

#	Part	Qty	RM each	RM
The brain				
1	**Raspberry Pi 4 Model B, 4 GB** Runs the robot program, its own server and the AI agent	1	290	290
2	**microSD card 32 GB, A1 / Class 10 (SanDisk, Samsung)** Holds Raspberry Pi OS and the robot software	1	25	25
3	**Heatsink set or small fan for the Pi 4** The Pi works hard (camera + AI); keeps it from slowing down	1	15	15
4	**Official Pi 4 USB-C power supply 15 W** For the desk while you set up and test; the robot runs on batteries	1	40	40
Power				
5	**Waveshare UPS Module 3S (3 × 18650, 5 V 5 A out)** Battery for the Pi, sensors and servos; charges while running; has a battery gauge	1	130	130
6	**18650 Li-ion cells, flat top, 2500-3000 mAh (Samsung 25R/30Q, LG HG2)** 3 for the UPS module + 2 for the motors. Avoid fake '9900 mAh' cells	5	18	90
7	**2 × 18650 battery holder with on/off switch and leads** Separate motor battery (7.4 V): motor spikes never reset the Pi	1	8	8
8	**12.6 V 2 A charger, 5.5 × 2.1 mm plug (if the UPS module has none)** Charges the UPS module's 3 cells	1	20	20
9	**2-slot USB 18650 charger** Charges the 2 motor cells	1	20	20
10	**Short USB-C to USB-C cable (3 A, 20-30 cm)** UPS module 5 V output → Pi power input	1	12	12
Moving				
11	**4WD smart car chassis kit: 2 acrylic decks, 4 TT motors (3-6 V), 4 wheels, brackets, screws** The body. Choose a kit whose motors have a shaft on both sides (for the encoder discs)	1	60	60
12	**L298N dual H-bridge motor driver** Turns the Pi's small signals into motor power	1	10	10
13	**PCA9685 16-channel PWM driver (I^2C)** Smooth motor speed and the two servo signals	1	18	18
14	**SG90 micro servo** Pan (left/right) and tilt (up/down) of the camera	2	7	14
15	**Pan/tilt bracket for SG90** Holds the two servos and the camera	1	8	8
16	**Speed encoder module (LM393 slot sensor) + 20-slot disc** Counts wheel turns: 'go forward 2 m' stops at 2 m	2	5	10
Sensing				
17	**HC-SR04 ultrasonic distance sensor** Left, centre, right: stops before obstacles and steers round them	3	5	15

#	Part	Qty	RM each	RM
18	**HC-SR04 mounting bracket** Holds the sensors at 45° left, straight, 45° right	3	2	6
19	**MPU6050 motion sensor (GY-521 board)** Knows how far the robot has turned	1	10	10
20	**BME680 climate sensor board** Temperature, humidity, pressure and VOC air quality	1	45	45
21	**MQ-135 gas sensor module** CO2 / gas level	1	10	10
22	**ADS1115 16-bit ADC board** Reads the MQ-135's voltage (the Pi has no analog input)	1	15	15
Camera and sound				
23	**Raspberry Pi Camera Module v2 (8 MP)** Photos and live video. Budget option: OV5647 5 MP camera (~RM 35)	1	120	120
24	**Camera ribbon for Pi 4, 15-pin, 30-50 cm** The short ribbon in the box cannot reach the pan/tilt head	1	8	8
25	**USB mini microphone** Hears talking, alarms and machines	1	15	15
26	**USB sound card (3.5 mm output)** Clean audio out for the speaker (the Pi 4 jack is noisy)	1	10	10
27	**PAM8403 amplifier board** Makes the voice loud enough	1	4	4
28	**Speaker 3 W 4 Ω, about 40 mm** The robot's voice	1	6	6
29	**3.5 mm male audio cable (you will cut one end)** Sound card → amplifier	1	4	4
Wiring and mounting				
30	**Resistors: 1 kΩ x 3, 2 kΩ x 3, 10 kΩ x 1, 20 kΩ x 1 (buy a resistor kit)** Voltage dividers that protect the Pi's 3.3 V pins	1	8	8
31	**Perfboard / stripboard about 7 × 9 cm** Your 'hub board' (one spare for mistakes)	2	3	6
32	**Jumper wires: F-F, M-F, M-M, 40 of each** Most connections plug in, no soldering	1	15	15
33	**Male + female pin header strips** Plug-in points on the hub board	1	5	5
34	**Silicone wire 22 AWG, red + black, 2 m each** Motor and battery wiring	1	8	8
35	**M3 brass standoffs 35-40 mm + M2.5 / M3 screw and nut kit** Second deck and board mounting	1	18	18
36	**Heat-shrink, zip ties, double-sided foam tape** Tidy, safe, nothing rattles loose	1	15	15
	Total for the robot (estimate)			**≈ 1113**

Tools (if you do not have them)

Tool	RM
Soldering iron kit with solder (60/40 or lead-free) and stand Motor leads, hub board, header pins	50
Digital multimeter Check every voltage BEFORE connecting the Pi	40
Small screwdriver set (Phillips + flat) Screw terminals and chassis	15
Wire stripper / side cutter Clean wire ends	15
Hot glue gun Fix the microphone, cables and the encoder sensors	20

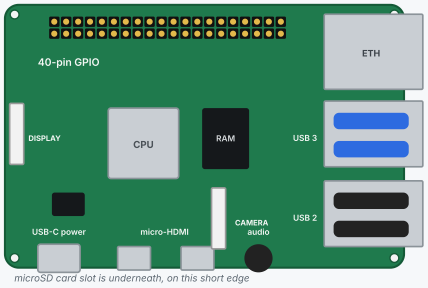
Buying tips

- Buy **one spare** HC-SR04 and one spare jumper-wire set: cheap parts sometimes arrive faulty.
- **18650 cells:** buy known brands from a trusted seller. Real cells are 2500-3500 mAh; anything claiming "9900 mAh" is fake and unsafe. Flat-top cells fit most holders.
- Check the **chassis kit** photo: TT motors with a shaft on **both** sides, and encoder discs included.
- The **Pi Camera v2** and the Pi 4 use the same 15-pin ribbon; order a 30-50 cm ribbon with it.
- No 2 kΩ resistors? Two 1 kΩ in series make 2 kΩ.
- Already have a Pi 4 power supply? Keep it: it is the safest way to power the Pi on your desk while you set up.

Budget: about **RM 1,113** for the robot, plus about RM 140 of tools. The budget camera (OV5647, ~RM 35) saves about RM 85; everything else in the software works the same.

3 Meet the parts

What each part looks like, what it does, which pins you will use and one tip. The pictures are drawings of the common versions; your boards may differ slightly in colour or pin order, so always read the labels printed on the board.

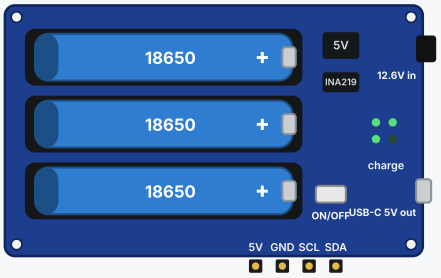


Raspberry Pi 4

The robot's computer: drives, thinks, talks to the phone.

Connections: 40-pin GPIO header (top edge) · USB-C power · CAMERA port · USB ports for the mic and sound card

💡 4 GB is enough. Put the microSD card in the slot underneath on the short edge.

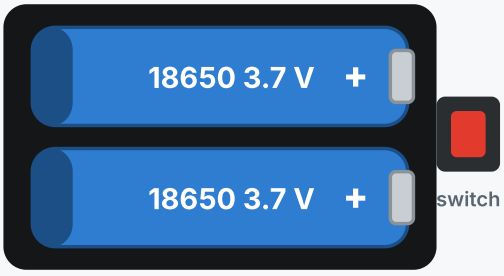


UPS Module 3S

Holds 3 × 18650 cells and gives a steady 5 V 5 A to the Pi.

Connections: USB-C 5 V out → Pi · 5V/GND/SCL/SDA header → hub (gauge) and PCA9685 V+ · 12.6 V charger input

💡 Insert cells with + to the + mark. Switch off before wiring.

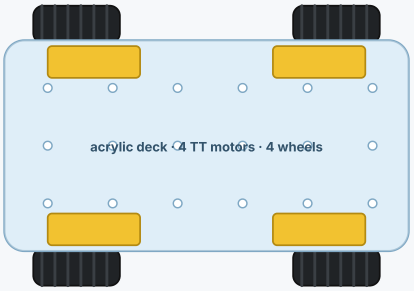


Motor battery 2 × 18650

Feeds only the motors, through the L298N.

Connections: Red lead + → L298N +12V · black lead – → L298N GND

💡 7.4 V nominal, 8.4 V full. Use its switch as the motor power switch.

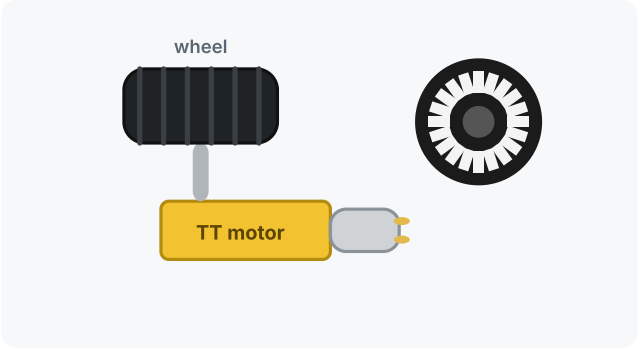


4WD chassis kit

Two acrylic decks, four TT gear motors and wheels.

Connections: Motors under the bottom deck · second deck on standoffs

💡 Check the motors have a shaft on both sides for the encoder discs.

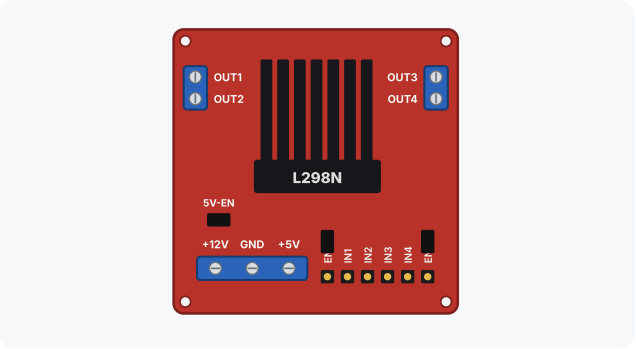


TT gear motor + encoder disc

3-6 V geared motor; the 20-slot disc tells how far it turned.

Connections: 2 solder tabs per motor (red / black wires)

💡 Solder wires before you mount the motors; it is hard afterwards.

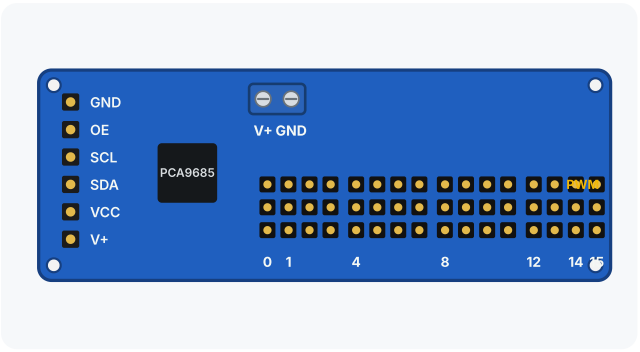


L298N motor driver

Switches battery power to the motors forwards or backwards.

Connections: +12V / GND from the motor pack · OUT1-OUT4 to motors · IN1-IN4 from the Pi · ENA/ENB from the PCA9685

💡 Remove the ENA and ENB jumpers. Keep the 5V-EN jumper.

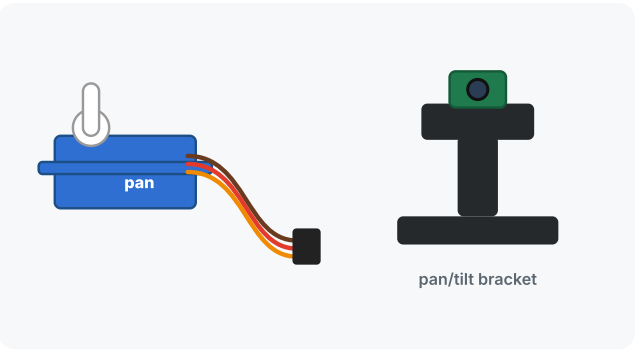


PCA9685 PWM driver

16 precise PWM outputs over two I²C wires.

Connections: GND OE SCL SDA VCC V+ header · V+ screw terminal · channels 0-15 (signal / V+ / GND)

💡 VCC = 3.3 V (logic). V+ = 5 V for the servos.

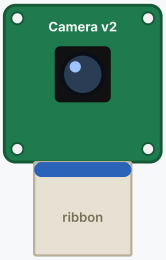


SG90 servos + pan/tilt bracket

Turn the camera left/right and up/down.

Connections: 3-wire plug: brown GND · red 5 V · orange signal

💡 Never turn a powered servo by hand; it strips the gears.

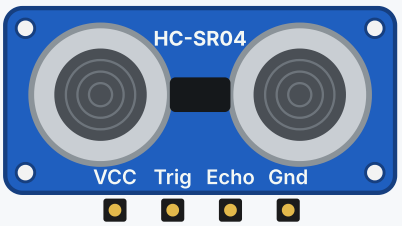


Pi Camera Module v2

8 MP photos and live video for the AI.

Connections: 15-pin ribbon → CAMERA port on the Pi

💡 Lift the connector latch, insert the ribbon fully, press the latch down.

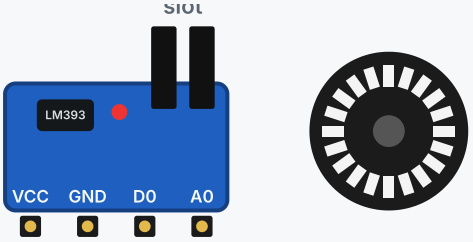


HC-SR04 ultrasonic

Measures distance with sound, 2 cm to 4 m.

Connections: VCC 5V · Trig (from Pi) · Echo (5 V out! needs a divider) · Gnd

💡 Buy 3 + a spare; they are cheap and sometimes arrive dead.

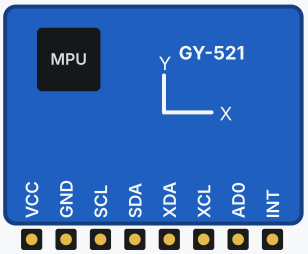


Speed encoder (slot sensor)

Gives one pulse each time a slot of the disc passes.

Connections: VCC 3.3V · GND · D0 (digital, to the Pi) · A0 (not used)

💡 The disc must spin inside the slot without touching it.

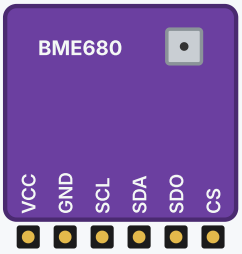


MPU6050 (GY-521)

Gyroscope + accelerometer: the robot's sense of turning.

Connections: VCC · GND · SCL · SDA (the others are not used)

💡 Mount it flat, near the middle of the chassis, with foam tape.

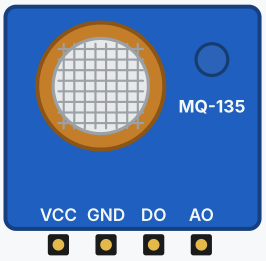


BME680

Temperature, humidity, pressure and VOC gas.

Connections: VCC · GND · SCL · SDA (SDO open = address 0x76)

💡 Keep it away from the Pi and the MQ-135: they are warm.

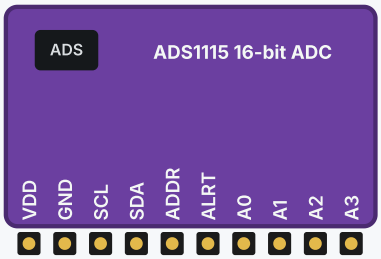


MQ-135 gas sensor

Heated sensor whose voltage rises with CO2 and gases.

Connections: VCC 5V · GND · AO (analog, via divider) · DO (not used)

💡 Burn it in for 24-48 h the first time. It gets hot: no plastic touching it.



ADS1115 ADC

Converts the MQ-135 voltage into a number for the Pi.

Connections: VDD 3.3V · GND · SCL · SDA · ADDR → GND · A0 = gas input

💡 Powered from 3.3 V so it is safe on the Pi's I²C bus.



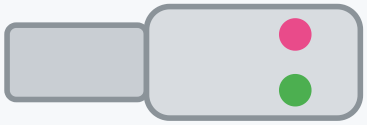
USB microphone

USB microphone

Hears voices, alarms and machines.

Connections: Any USB port

💡 Point it forward; fix it with hot glue.



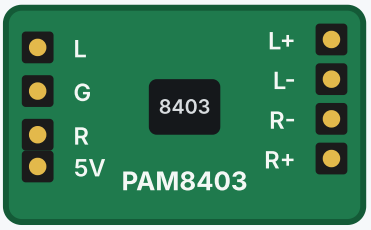
USB sound card

USB sound card

Clean audio output.

Connections: Any USB port · green jack = output

💡 Plug the 3.5 mm cable into the green jack.



PAM8403 amplifier

Small 3 W amplifier for the speaker.

Connections: 5V/GND power · L G R input · L+ L− output

💡 Use only the left channel (L+ / L−).



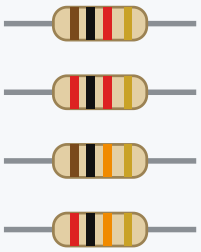
3 W 4 Ω speaker

Speaker 3 W 4 Ω

The robot's voice.

Connections: 2 wires to L+ and L−

💡 Face it outwards at the back of the top deck.

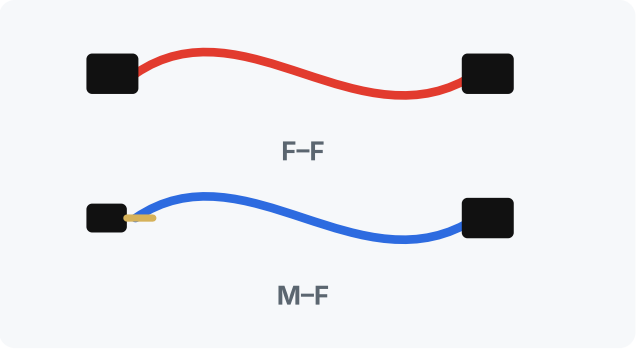


Resistors

Voltage dividers protecting the Pi.

Connections: No polarity: either way round

💡 No 2 kΩ? Two 1 kΩ in series = 2 kΩ.

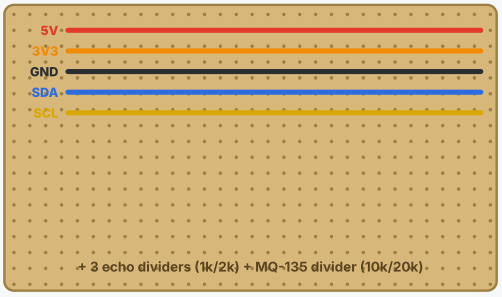


Jumper wires

Plug-in connections.

Connections: F = socket (fits on a pin) · M = pin

💡 Use colours: red 5 V, orange 3.3 V, black GND.



Hub board (you make it)

One place where power, the I²C bus and all dividers meet.

Connections: 5V · 3V3 · GND · SDA · SCL rails + 4 dividers

💡 Step 5 shows how to build it.

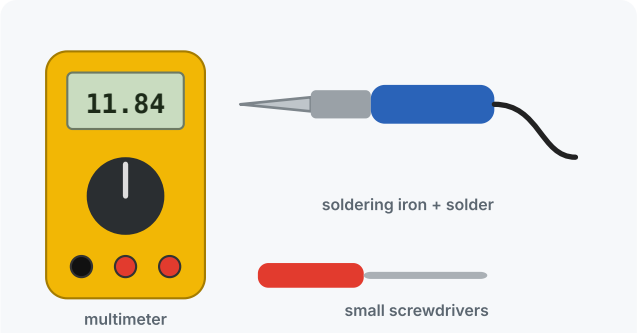


Standoffs and screws

Hold the second deck and the boards.

Connections: M3 for the deck, M2.5 for the Pi

💡 Nylon washers stop shorts under boards.



Tools

What you need on the desk.

Connections: Soldering iron, multimeter, screwdrivers, stripper, hot glue

💡 Measure, then connect. Always.

4 Safety first

Lithium batteries store a lot of energy.

- Never short the + and − of a cell or pack (no loose wires, no metal tools across terminals).
- Check polarity three times before inserting a cell: + goes to the + mark.
- Charge only with a proper Li-ion charger, never unattended, on a non-flammable surface.
- Do not use cells with dents, torn wrapping or that get hot; recycle them.
- Switch both batteries off before changing any wire.

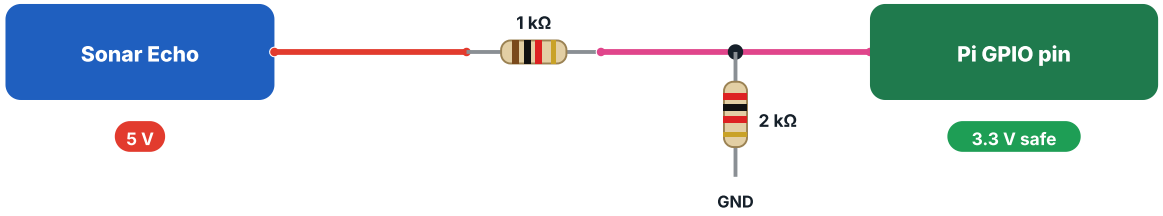
Protect the Raspberry Pi.

- Its GPIO pins are **3.3 V only**. 5 V on a GPIO pin damages it instantly, so every 5 V output goes through a divider.
- Never power motors or servos from the Pi's pins.
- All grounds (Pi, UPS, motor pack, L298N) must be joined: the **common ground**.
- Measure each power rail with the multimeter before plugging a board in.
- Touch something metal before handling boards (static electricity).

Moving parts.

- Test motors with the robot on a box, **wheels in the air**.
- Keep fingers and hair away from the wheels and the pan/tilt head.
- The MQ-135 heater and the L298N heatsink get hot.

The two rules you will use most



A voltage divider: 5 V × 2 kΩ / (1 kΩ + 2 kΩ) = 3.33 V, safe for the Pi.

And: **measure, then connect**. Put the multimeter on DC volts, black probe on GND, red probe on the rail. 5 V rail: 4.9-5.2 V. 3.3 V rail: 3.2-3.4 V. Motor pack: 7.0-8.4 V.

5 The pin map

Every Raspberry Pi pin the robot uses. Physical pin numbers (1-40) are what you count on the header; the GPIO names are what the software uses (they are already set in `edge/config.yaml`).



Figure 2 - Raspberry Pi 4 header, top view, USB ports pointing down the page.

> Everything else does not touch the header: the camera uses its ribbon port, the microphone and sound card use USB, the servos and motor speed use the PCA9685, and the gas sensor goes through the ADS1115.

6 Build it in 13 steps

Each step lists what you need, what to do, a picture, and a check. Do them in order.

1 Prepare the Raspberry Pi (software first)

You need: Pi 4, microSD card, Pi power supply, your PC

1. On the PC install **Raspberry Pi Imager**. Choose *Raspberry Pi 4* → *Raspberry Pi OS (64-bit)* → your microSD card.
2. In the settings (gear / "Edit settings"): hostname `robot01`, a username and password, **enable SSH**, and add a Wi-Fi. Your **phone's hotspot** is the best one: the robot then has internet for the AI wherever you go.
3. Write the card, put it in the Pi (slot underneath), power the Pi with its USB-C supply. Wait 2 minutes.
4. On the PC, double-click `pack_robot_for_pi.bat` in the project folder. It makes `raqib-robot.zip`.
5. Copy the zip to the Pi (WinSCP to `robot01.local`, or a USB stick) and run on the Pi:

```
unzip raqib-robot.zip -d ~/raqib
cd ~/raqib && bash edge/scripts/install_pi.sh
sudo reboot
```

The installer sets real-hardware mode, starts the robot and its server at every power-on, and makes the robot's own Wi-Fi **Raqib-Robot** (password `raqib1234`) when no known Wi-Fi is around.

Check before you continue

☐ The Pi boots and you can log in (SSH or keyboard + screen)

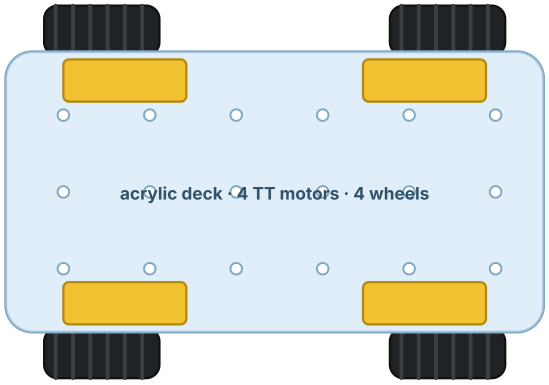
☐ The installer ended with 'Done'

☐ The phone app finds the robot ('Find automatically'); Robot health shows every part red: correct, nothing is wired yet

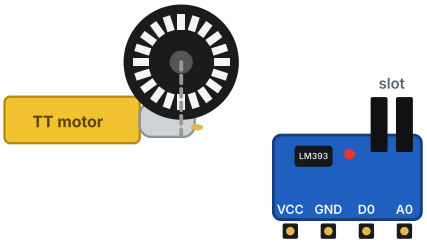
2 Chassis, motors and wheel encoders

You need: chassis kit, 4 TT motors, wheels, 2 encoder discs + modules, wire, soldering iron

1. **Solder** a red and a black 20 cm wire to the two tabs of each motor. Twist each pair.
2. Fix the 4 motors under the bottom deck with the kit's brackets and long screws, all wires towards the middle.
3. Push an **encoder disc** onto the inner shaft of the two **front** motors.
4. Mount each encoder module so its disc spins inside the slot without touching (small bracket, standoff or hot glue).
5. Push the wheels onto the outer shafts.
6. Join the two motors of each side: red to red, black to black (front and rear work as one).

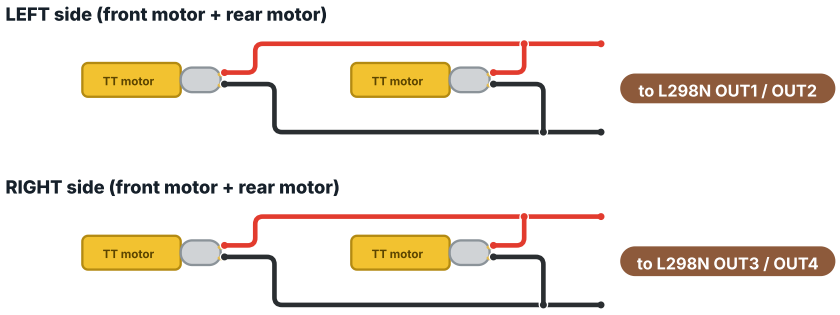


1. Screw the 4 motors under the bottom deck with the kit's brackets



2. Disc on the motor's inner shaft, spinning inside the slot of the encoder module

Motors under the bottom deck; the encoder disc spins in the sensor's slot.



Join red to red and black to black: the two motors of a side then work as one. A side that turns the wrong way: swap its two wires at the L298N.

Each side's two motors in parallel.

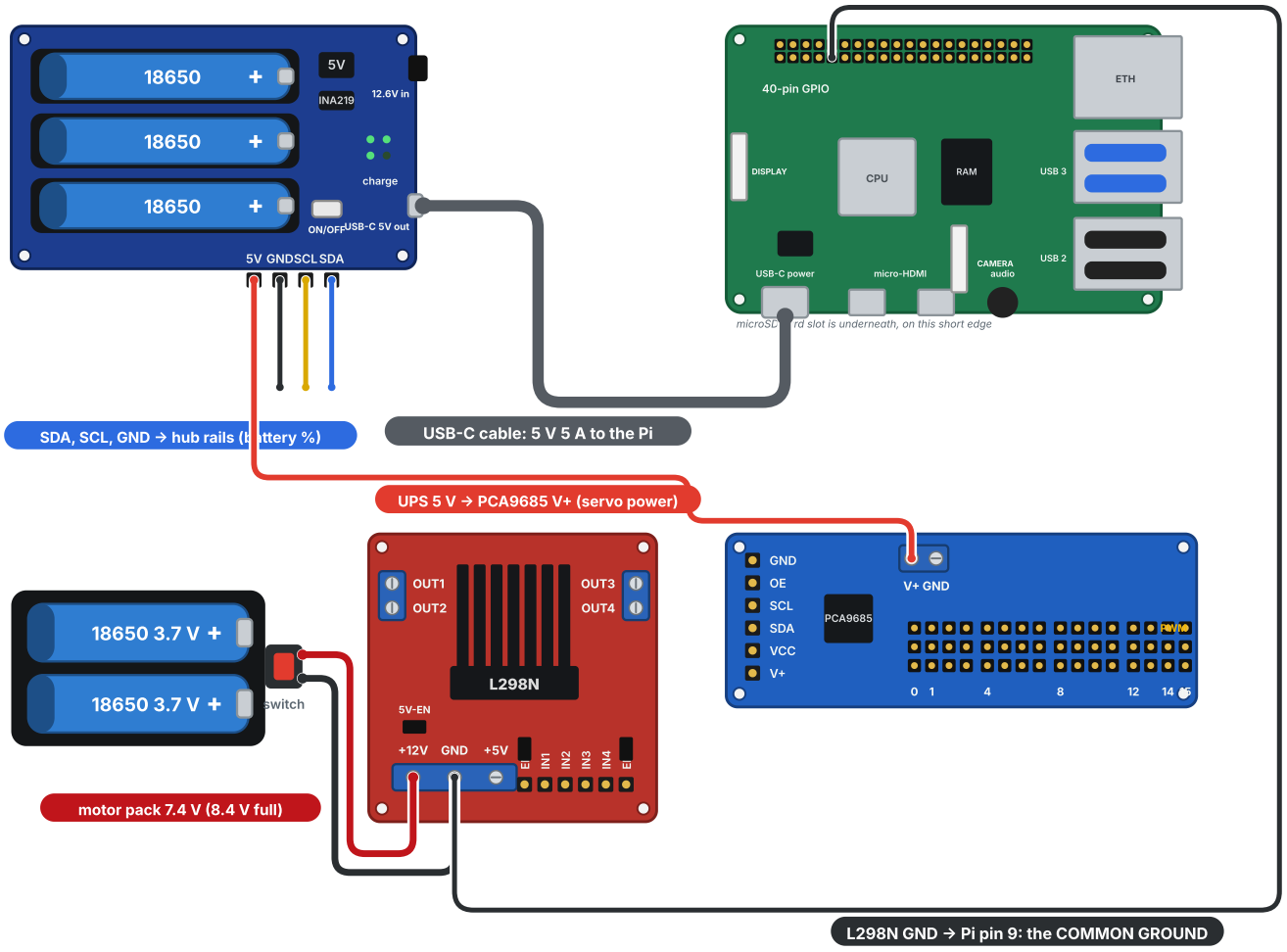
Check before you continue

- ☐ Every wheel turns freely by hand
- ☐ Each encoder disc passes through its slot without rubbing
- ☐ Left pair and right pair each have one red + one black wire to go to the L298N

3 Power: two batteries and the common ground

You need: UPS Module 3S, 3 + 2 cells, motor battery holder, L298N, USB-C cable, multimeter

1. With all switches OFF, insert the 3 cells into the UPS module and the 2 cells into the motor holder. **Check polarity.**
2. Fix the motor holder under the bottom deck and the L298N on the bottom deck (screws or foam tape).
3. Motor holder red → L298N **+12V**, black → L298N **GND**. Tighten the screw terminals.
4. Switch the motor pack on and measure L298N +12V to GND: **7-8.4 V**. Switch off again.
5. Switch the UPS on and measure its 5V header pin to GND: **about 5.1 V**. Switch off.
6. Only now: USB-C cable from the UPS module to the Pi's power input (replaces the desk power supply).
7. A wire from L298N **GND** to Pi **pin 9** (GND): the common ground (you will connect it in step 7).



Two batteries: the UPS module (3 × 18650) feeds the Pi, sensors and servos; a separate 2 × 18650 pack feeds the motors, so motor current spikes never reset the Pi. All grounds are joined.

Figure 3 - power wiring.

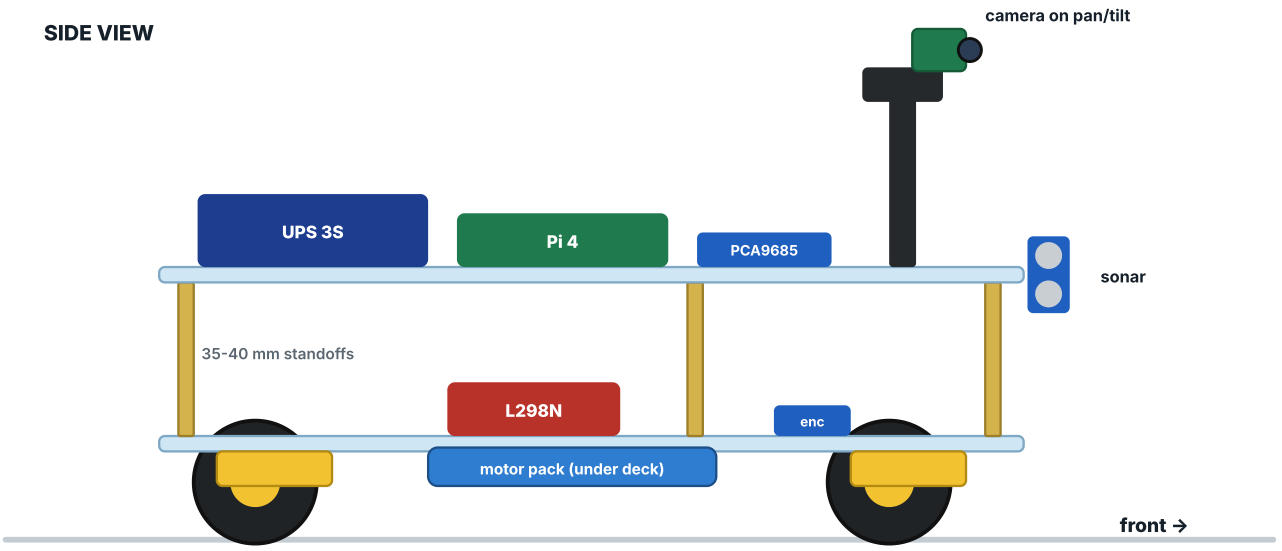
Check before you continue

- ☐ Motor pack reads 7-8.4 V at the L298N
- ☐ UPS 5 V reads 4.9-5.2 V
- ☐ The Pi boots from the UPS module alone
- ☐ Charging: the 12.6 V charger plugs into the UPS module; the motor cells go in the USB charger

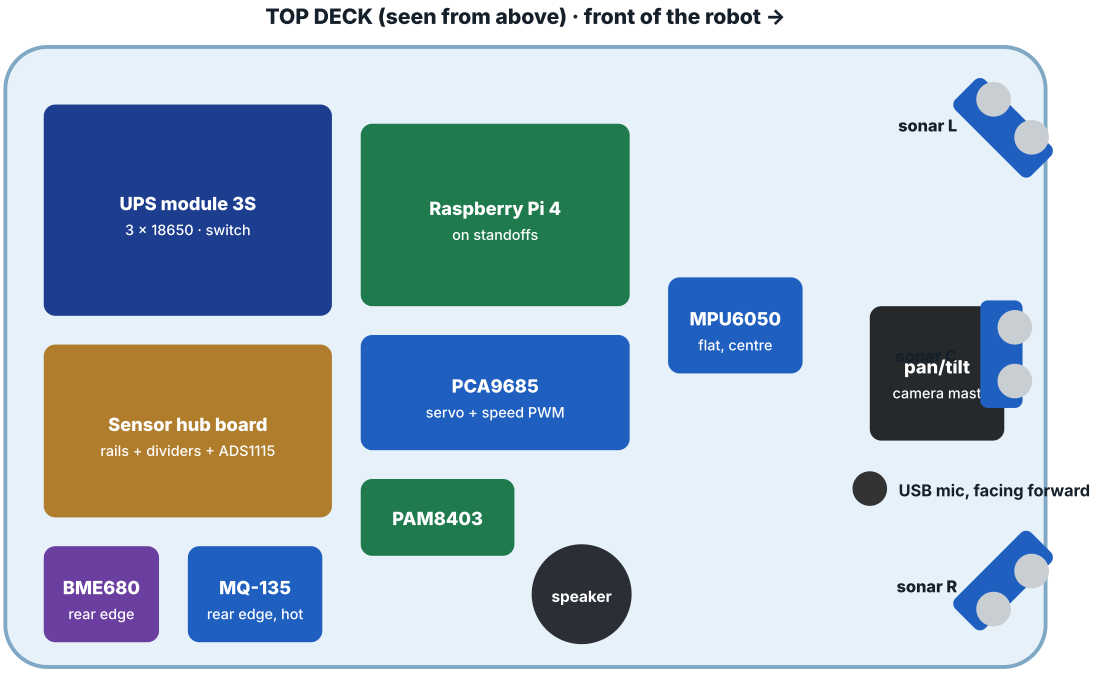
4 Mount the boards

You need: standoffs, screws, foam tape, top deck

Follow the layout: heavy batteries low and central, warm parts (Pi, MQ-135) away from the BME680, sonars on the front edge, the pan/tilt mast at the front, the microphone facing forward.



Side view: two decks on 35-40 mm standoffs.



Heat: keep BME680 and MQ-135 away from the Pi CPU. Weight: batteries low and central.

Top deck layout.

Check before you continue

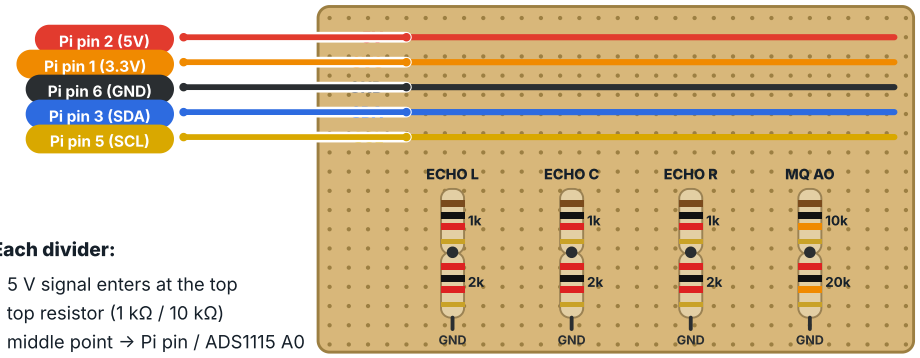
- ☐ Nothing touches the wheels
- ☐ No board's underside touches metal (use nylon washers)
- ☐ The top deck can still come off to reach the L298N

5 Build the hub board

You need: perfboard, resistors (3 × 1k, 3 × 2k, 1 × 10k, 1 × 20k), header strips, wire, soldering iron

The hub board is the heart of the wiring: five rails that every sensor plugs into, and the four voltage dividers.

1. Solder five long rails (bare wire or the board's strips): **5V, 3V3, GND, SDA, SCL**. Label them with a marker.
2. Solder a female header strip along each rail so jumper wires can plug in.
3. Solder the three sonar dividers (1 kΩ on top, 2 kΩ to GND) and the MQ-135 divider (10 kΩ on top, 20 kΩ to GND). Put a header pin on the input and on the middle point of each.
4. Connect the rails to the Pi with F-F jumpers: pin 2 → 5V, pin 1 → 3V3, pin 6 → GND, pin 3 → SDA, pin 5 → SCL.



Each divider:

- 5 V signal enters at the top
- top resistor (1 kΩ / 10 kΩ)
- middle point → Pi pin / ADS1115 A0
- bottom resistor (2 kΩ / 20 kΩ) → GND
- 5 V × 2/3 = 3.3 V: safe

Solder the rails as bare wire along the strips;
every module plugs into the rails with jumper wires.

Figure 4 - the hub board: rails on top, dividers below.

Check before you continue

- ☐ Multimeter, Pi on, nothing else plugged in: 5V rail ≈ 5 V, 3V3 rail ≈ 3.3 V
- ☐ No two rails are shorted (continuity mode: no beep between rails)
- ☐ Each divider: 5 V on its input reads ≈ 3.3 V at its middle point

6 I²C boards: PCA9685, MPU6050, BME680, ADS1115

You need: the four boards, F-F jumpers

Every I²C board gets four wires from the hub rails: **3V3, GND, SDA, SCL**. The UPS module's SDA/SCL/GND join the same rails so the app can show the battery %.

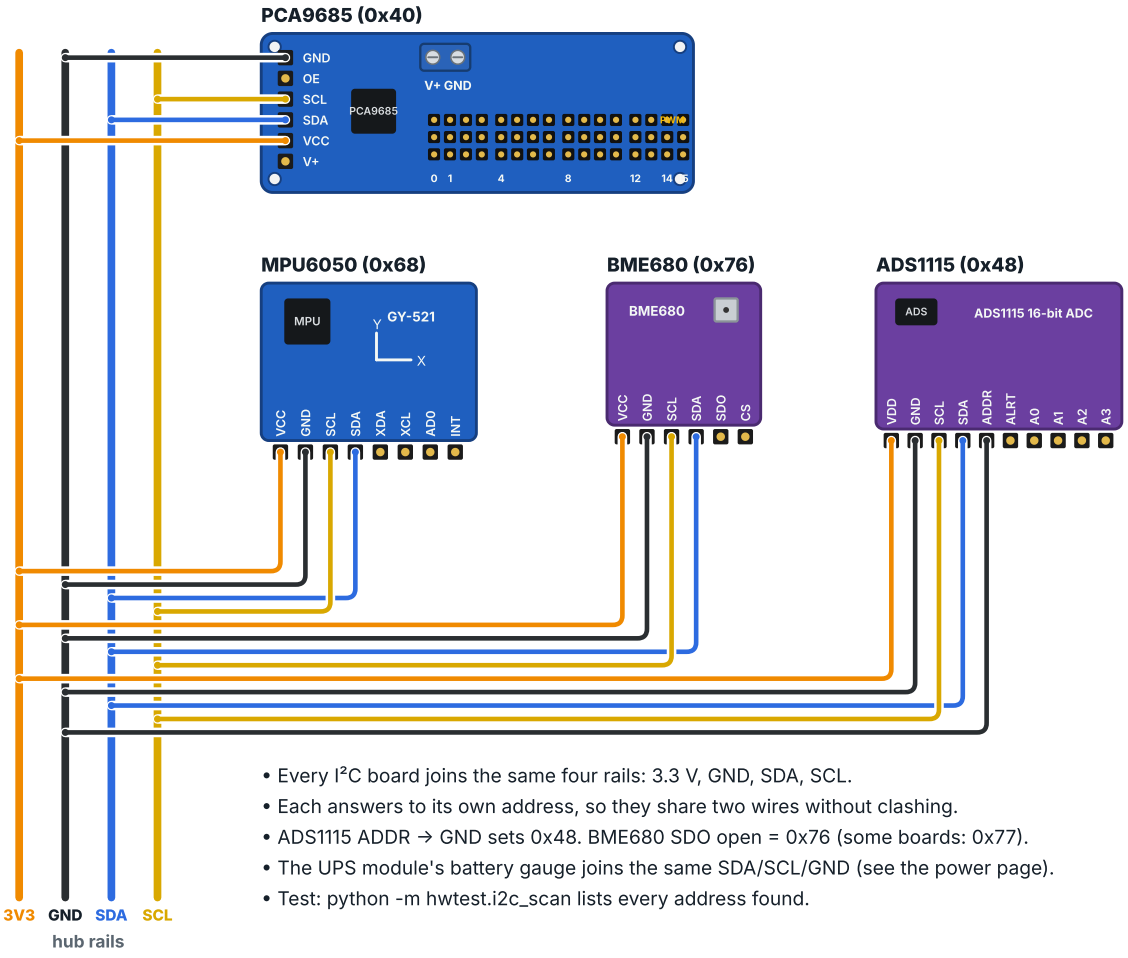


Figure 5 - the I²C bus.

```
sudo systemctl stop inspection-robot     # stop the robot program while you test
cd ~/raqib/edge && source .venv/bin/activate
python -m hwtest.i2c_scan
```

Check before you continue

☐ i2c_scan shows 0x40 (PCA9685), 0x48 (ADS1115), 0x68 (MPU6050), 0x76 or 0x77 (BME680) and the UPS gauge

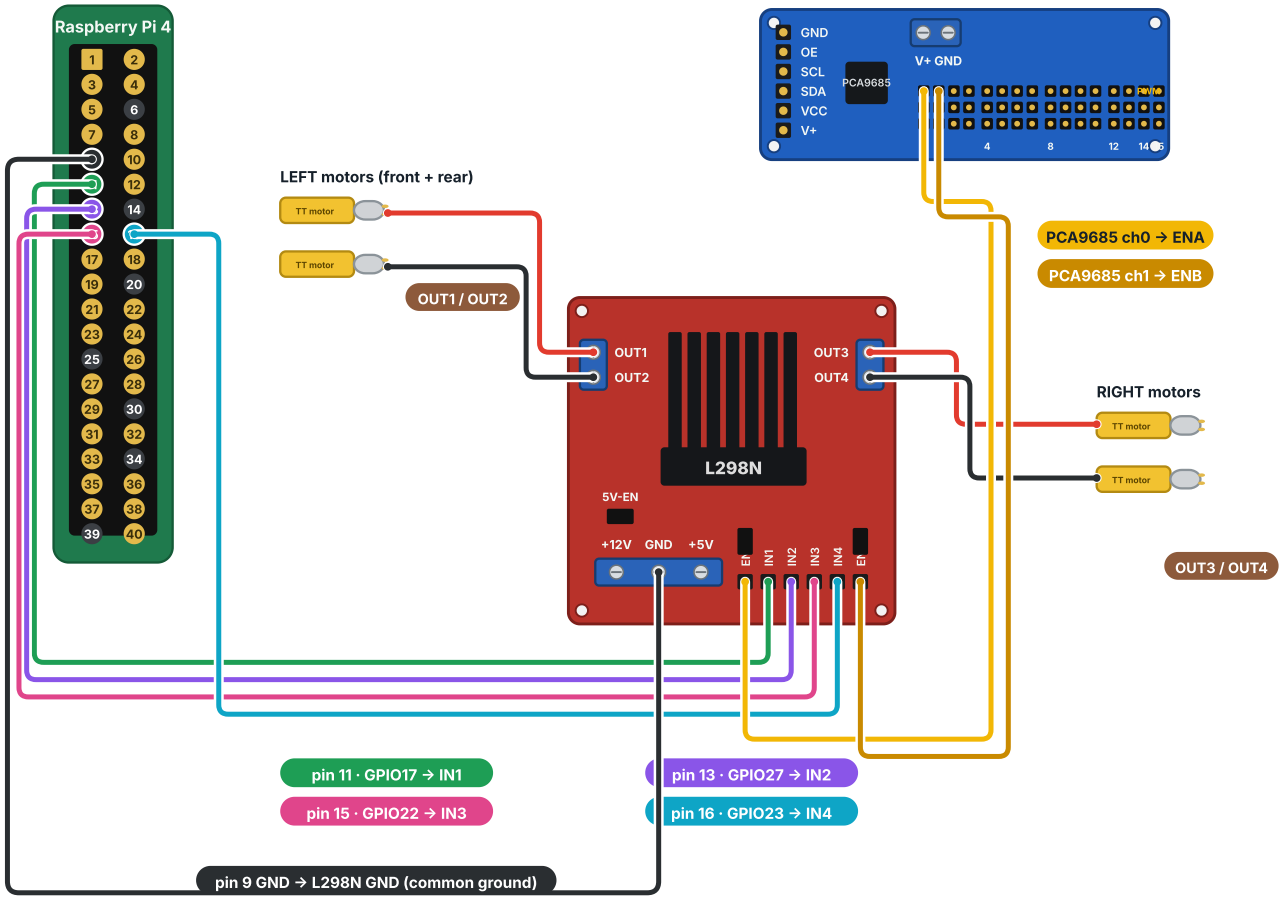
☐ BME680 at 0x77? set bme680_address: 0x77 in edge/config.yaml

☐ UPS gauge at another address (0x41 / 0x43)? set ups_ina219_address to it

7 Motors: L298N, PCA9685 and the Pi

You need: L298N (already powered), PCA9685, M-F / F-F jumpers

1. Pull the **ENA** and **ENB** jumpers off the L298N. Leave the 5V-EN jumper.
2. Pi pins 11, 13, 15, 16 → L298N IN1, IN2, IN3, IN4.
3. PCA9685 channel 0 signal → ENA, channel 1 signal → ENB.
4. Left motor pair → OUT1/OUT2, right motor pair → OUT3/OUT4.
5. L298N GND → Pi pin 9 (common ground).



- Remove the ENA and ENB jumpers (the PCA9685 sets the speed). Keep the 5V-EN jumper: the motor pack is under 12 V.
- +12V and GND terminals come from the motor pack (power page). Leave the +5V terminal unconnected.
- IN pins set the direction, ENA/ENB the speed. A side that runs backwards: swap its two motor wires.

Figure 6 - motor wiring.

```
python -m hwtest.test_motors          # robot on a box, wheels in the air!
```

Check before you continue

☐ Each side turns forwards when told 'forward'. Wrong way: swap that side's OUT wires (or invert: true in config)

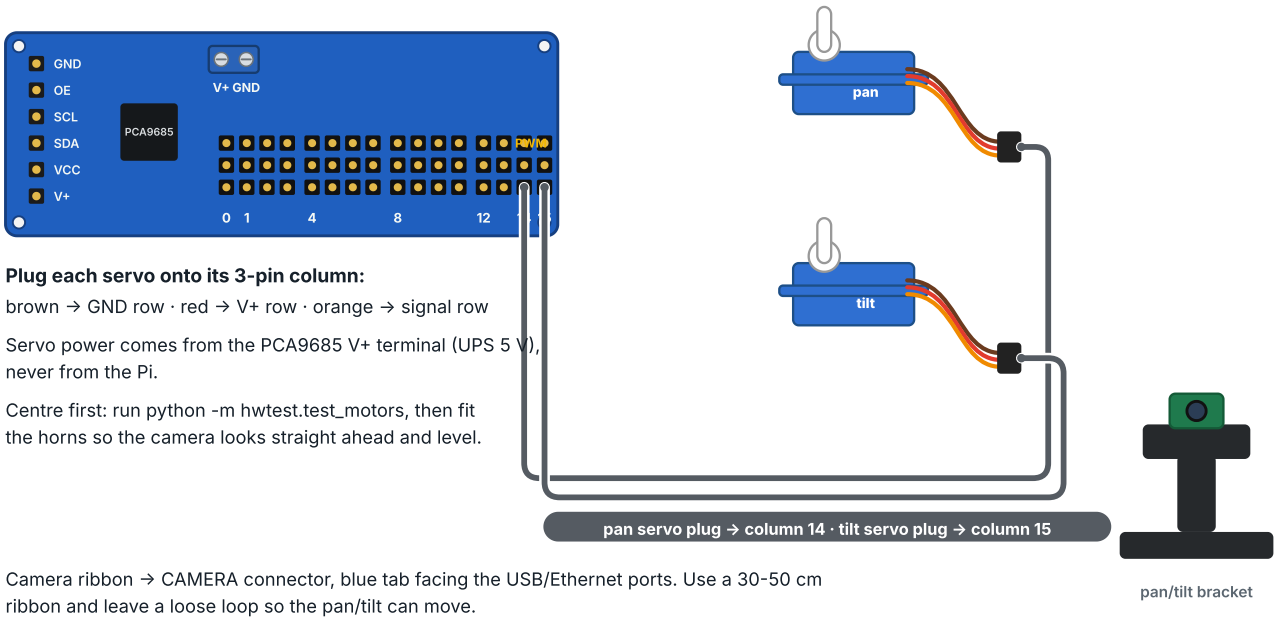
☐ Encoder ticks rise for both wheels (shown by the test)

☐ Floor test: python -m hwtest.test_motors --straight 5 drifts less than 20 cm

8 Pan/tilt servos and the camera

You need: 2 SG90, pan/tilt bracket, Pi Camera, long ribbon

1. Assemble the pan/tilt bracket with the two servos (follow the bracket's leaflet).
2. Plug the pan servo onto PCA9685 column 14 and the tilt servo onto column 15 (brown to GND).
3. UPS 5 V → PCA9685 V+ terminal (done in step 3).
4. Run the motor test once: the servos move to the centre. Now fit the horns so the camera faces straight ahead.
5. Screw the camera onto the tilt plate; ribbon to the Pi's CAMERA port, blue tab towards the USB/Ethernet ports, with a loose loop.



Plug each servo onto its 3-pin column:
brown → GND row · red → V+ row · orange → signal row

Servo power comes from the PCA9685 V+ terminal (UPS 5 V), never from the Pi.

Centre first: run python -m hwtest.test_motors, then fit the horns so the camera looks straight ahead and level.

Camera ribbon → CAMERA connector, blue tab facing the USB/Ethernet ports. Use a 30-50 cm ribbon and leave a loose loop so the pan/tilt can move.

Figure 7 - servos on the PCA9685.

Check before you continue

- ☐ Pan sweeps left/right, tilt up/down, no buzzing at rest
- ☐ The ribbon does not pull when the head turns fully

9 Three distance sensors

You need: 3 HC-SR04, brackets, F-F jumpers

1. Mount the sensors on the front edge: left one turned 45° left, centre straight, right one 45° right.
2. VCC → hub 5V, Gnd → hub GND.
3. Trig → Pi pin directly (3.3 V is enough to trigger).
4. Echo → its divider's input on the hub board; the divider's middle point → the Pi pin.

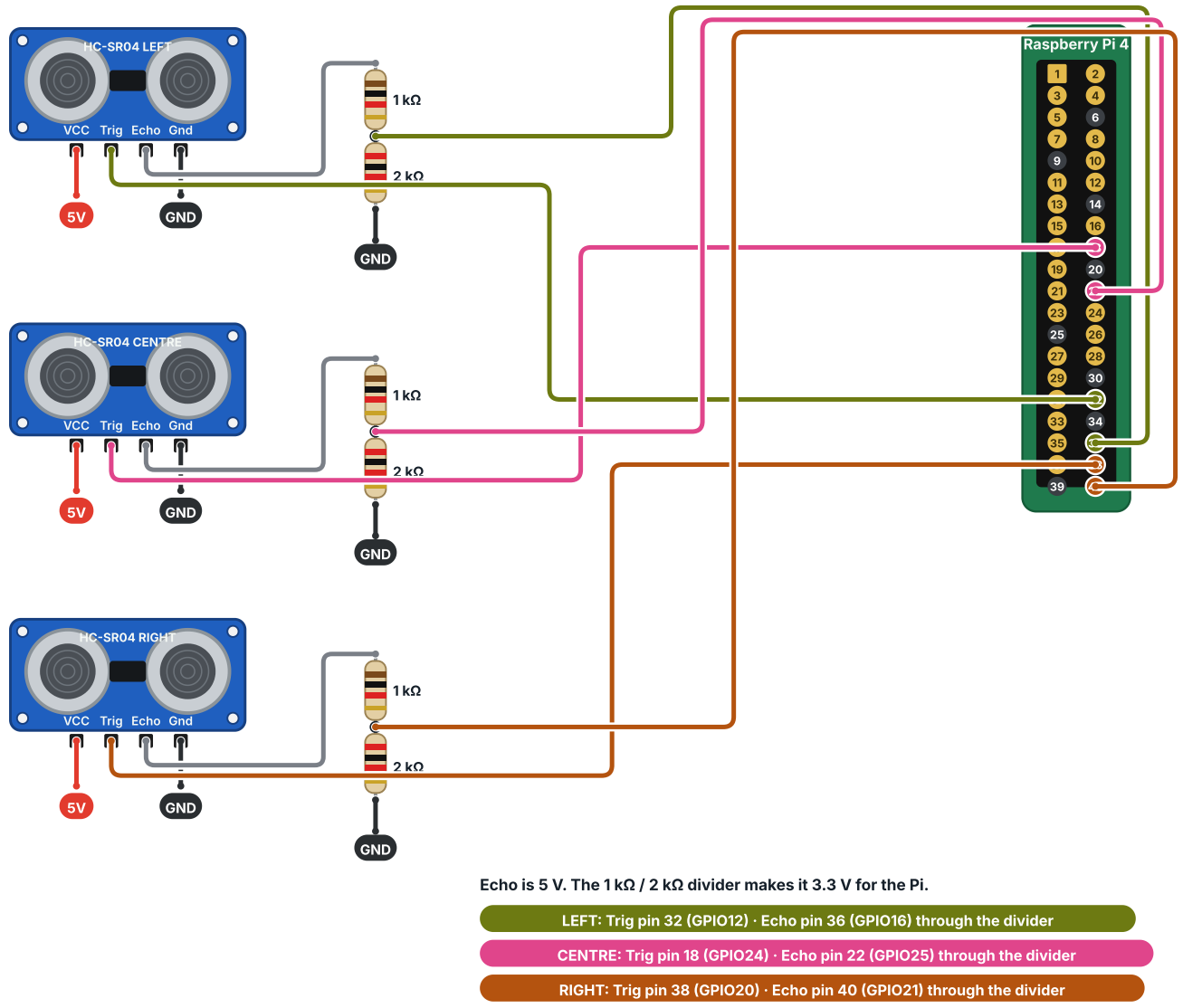


Figure 8 - ultrasonic sensors and their dividers.

```
python -m hwtest.test_sensors
```

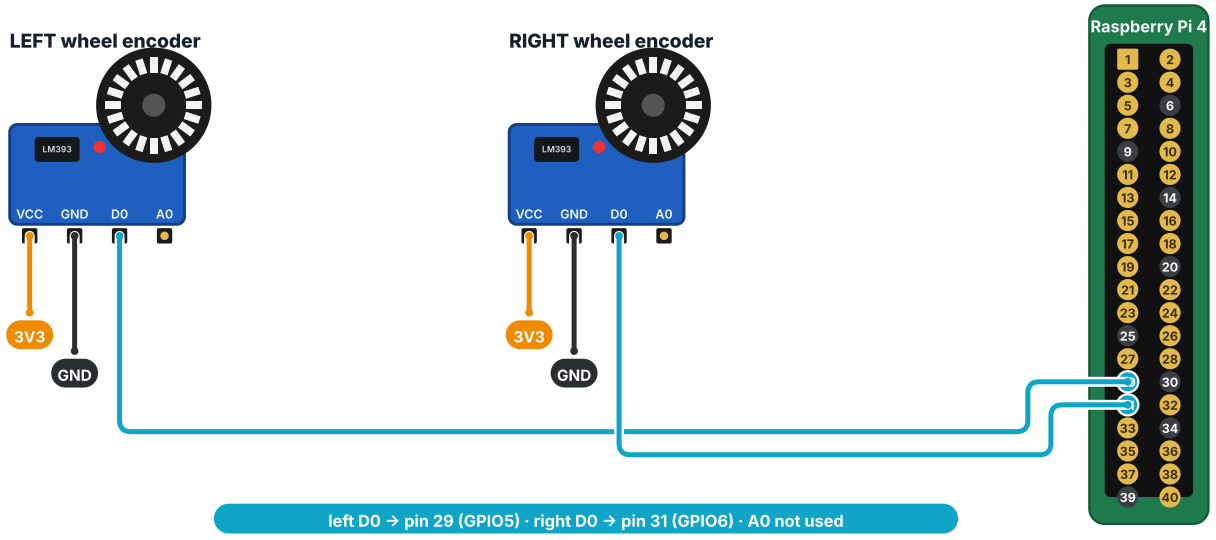
Check before you continue

- ☐ Each distance follows your hand (left, centre, right)
- ☐ A constant 400 cm or 0 means Trig/Echo swapped or a divider fault

10 Wheel encoders

You need: the two encoder modules from step 2, F-F jumpers

1. VCC → hub **3V3** (not 5 V), GND → hub GND.
2. Left D0 → Pi pin 29, right D0 → Pi pin 31.



Power the encoders from 3.3 V so their output is safe for the Pi. The disc must spin freely in the slot.

Figure 9 - encoders.

Check before you continue

- ☐ test_sensors: the 'enc' numbers rise when you turn each wheel by hand
- ☐ The module's LED blinks as slots pass

11 Gas and climate sensors

You need: MQ-135, ADS1115 (wired in step 6), the hub's 10k/20k divider

1. MQ-135 VCC → hub 5V, GND → hub GND.
2. MQ-135 AO → the 10 kΩ/20 kΩ divider input; divider middle → ADS1115 A0.
3. Mount the MQ-135 and BME680 at the back edge, away from the Pi.

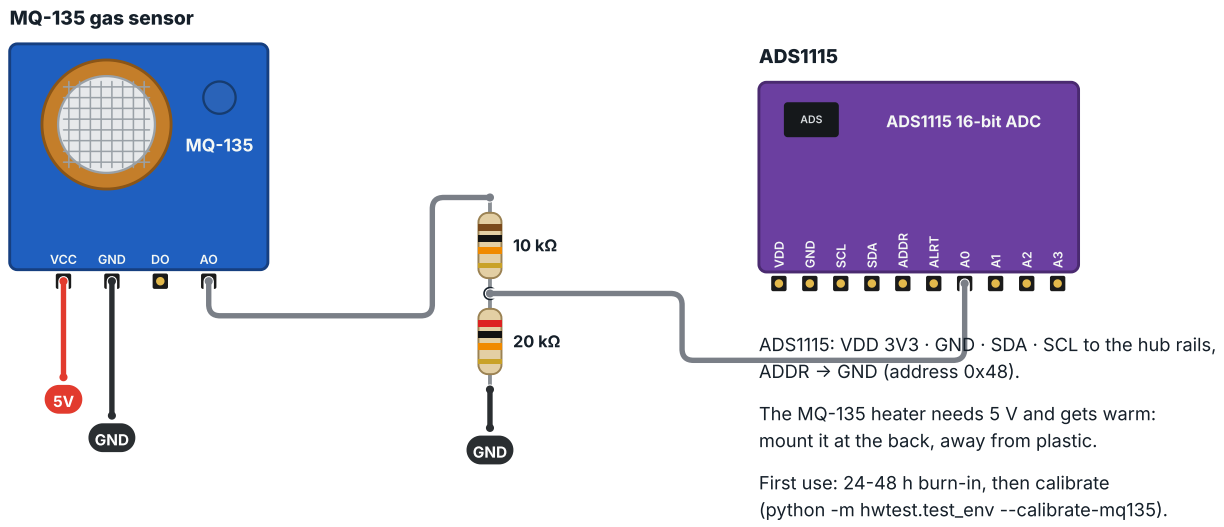


Figure 10 - MQ-135 through the divider into the ADS1115.

```
python -m hwtest.test_env                    # live readings
python -m hwtest.test_env --burn-in          # 5 min in clean air
python -m hwtest.test_env --calibrate-mq135  # outdoors, after 24 h
```

Check before you continue

- ☐ Temperature and humidity are close to a room thermometer
- ☐ Gas value settles after a few minutes of warm-up

12 Camera, microphone and speaker

You need: USB mic, USB sound card, PAM8403, speaker, 3.5 mm cable

1. Microphone and sound card into the Pi's USB ports.
2. Cut the 3.5 mm cable: the tip wire → PAM8403 L, the shield → G. PAM8403 5V/GND from the hub.
3. Speaker → PAM8403 L+ and L−.

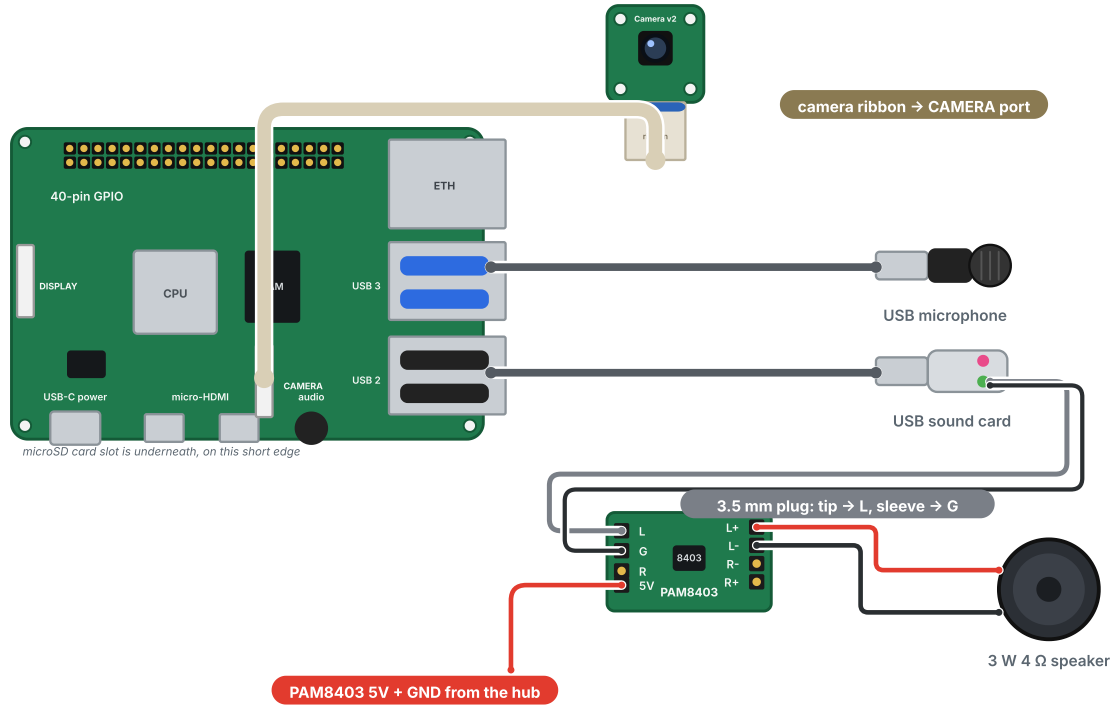


Figure 11 - camera, microphone and speaker.

```
python -m hwtest.test_camera_audio
```

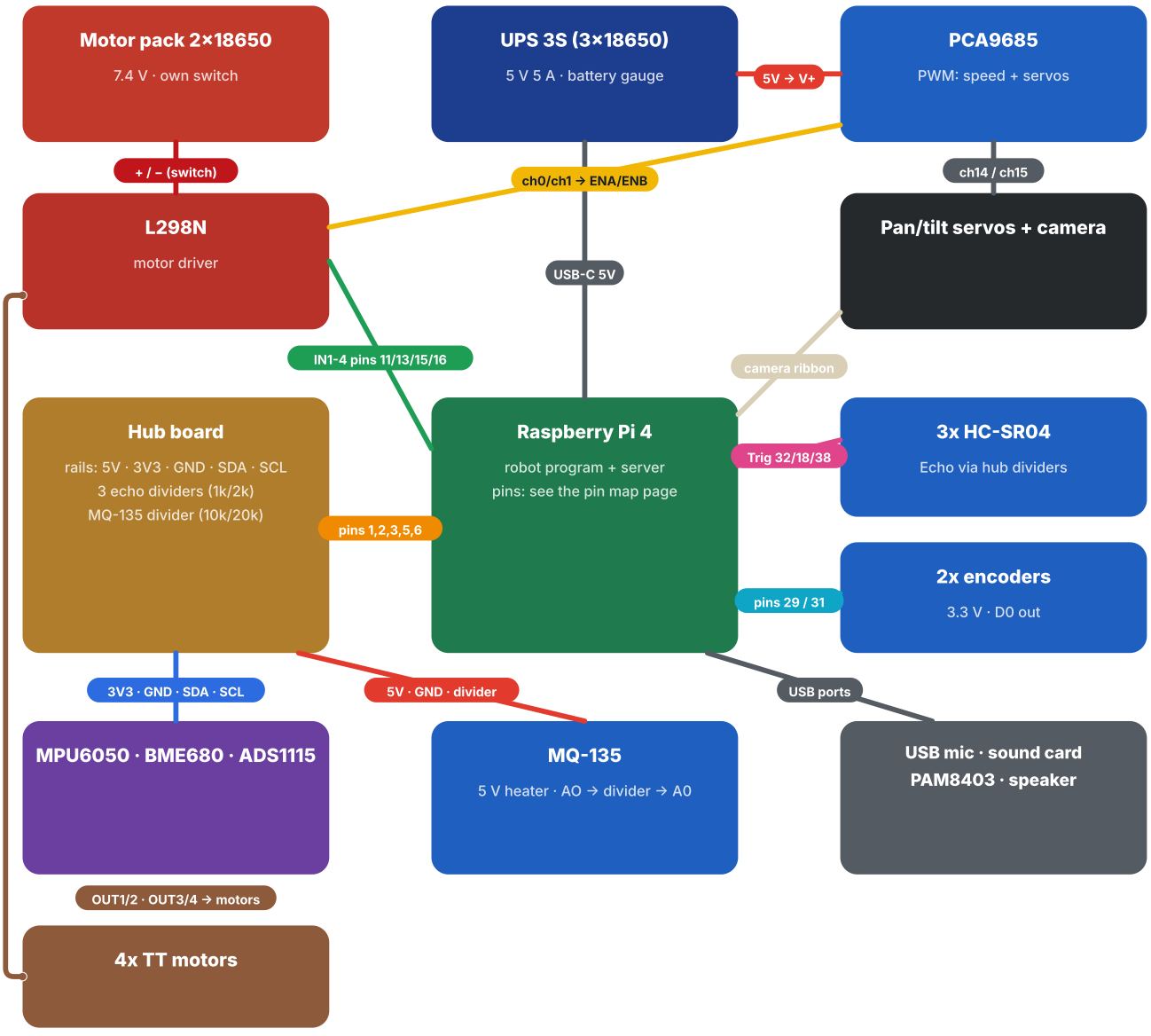
Check before you continue

- ☐ test_snapshot.jpg shows the room the right way up
- ☐ test_audio.wav has your voice
- ☐ The speaker says the test sentence

13 Everything together

You need: the finished robot, the phone with the Raqib app

1. Tidy the cables with zip ties; nothing near the wheels.
2. Start the robot program again: `sudo systemctl start inspection-robot` (or reboot).
3. Phone on the same Wi-Fi (or join **Raqib-Robot**). Open the app → **Find automatically** → sign in.
4. **More → AI settings**: paste your Gemini key once.
5. **More → Robot health**: every part should be green. Red? Tap it for the fix, then **Re-check parts**.
6. Drive to each room, read the position on the Home screen, and write them under `zones:` in `edge/config.yaml`.



Echo lines: sonar Echo → hub divider → Pi pins 36 / 22 / 40. Common ground: L298N GND → Pi pin 9.

Figure 12 - the complete harness on one page.

Check before you continue

- ☐ Every part green in Robot health
- ☐ STOP in the app stops the robot immediately
- ☐ A hand 30 cm in front of the moving robot stops it before 15 cm

7 Troubleshooting

What you see	Usual cause	Fix
i2c_scan finds nothing	I²C off, SDA/SCL swapped, or no 3.3 V on the rail	sudo raspi-config → Interface → I2C on; check pins 3/5; measure the 3V3 rail
One I²C board is missing	Loose jumper, or a different address	Re-seat its 4 wires; BME680 may be 0x77 → change the config
Wheels do not turn	ENA/ENB jumpers still on, motor pack off, no common ground	Remove the jumpers, switch the pack on, L298N GND → Pi pin 9
One side turns backwards	Motor wires reversed	Swap that side's two OUT wires, or set invert: true
Pi restarts when the motors start	Motors powered from the Pi's battery	Motors must use the separate 2 × 18650 pack
Sonar always 400 cm (or 0)	Trig/Echo swapped, no 5 V, wrong divider	Check the pin numbers and measure 5 V at the sensor
Encoder count stays 0	Disc not in the slot, D0 on the wrong pin, no 3.3 V	Watch the module's LED blink while turning the wheel
Servos jitter or do not move	No 5 V on the PCA9685 V+ terminal	UPS 5 V → V+, and GND shared
Camera not found	Ribbon reversed or not fully inserted	Blue tab towards USB ports; latch closed; run rpicam-hello
No sound	Wrong output device or no amp power	Check 5 V on the PAM8403; aplay -l lists the USB card
Gas value strange	Sensor still warming up, divider wrong	Wait 2-5 min; check 10k on top, 20k to GND
App cannot find the robot	Phone on another network	Same Wi-Fi or join Raqib-Robot; the robot says its address at start-up

Logs on the Pi: `journalctl -u inspection-robot -f` (robot) and `journalctl -u raqib-server -f` (server). The app's **Robot health** screen shows every part with a hint when it is red.

8 Final checklist before a demo

- ☐ Both batteries charged; spare charged cells in the bag
- ☐ Every part green in Robot health
- ☐ STOP in the app stops the robot; Reset is needed to move again
- ☐ A hand 30 cm in front of the moving robot stops it before 15 cm
- ☐ Stopping the server (sudo systemctl stop raqib-server) puts the robot in SAFE HOLD within about 10 s
- ☐ The robot starts at the Home spot, facing the same direction as when you measured the rooms
- ☐ The admin password in the app has been changed from admin123
- ☐ AI key entered in More → AI settings; "Test" says it works